METRIC
FF-B-171/20
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, ANGULAR CONTACT, COUNTERBORED OUTER RING, CONTACT ANGLE ABOVE 10° UP TO AND INCLUDING 22°, NON-SEPARABLE, ANSI/AFBMA DIMENSION SERIES 10.

The complete requirements for acquiring the bearings described. herein shall consist of this document and the latest issue of FF-B-171/GEN.

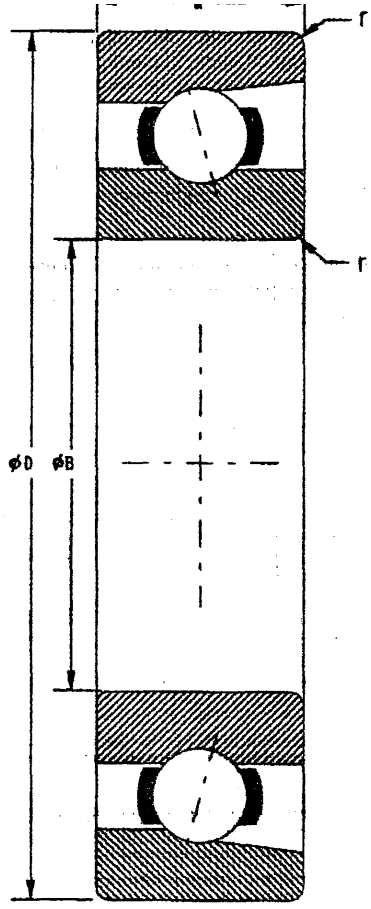


FIGURE 1.
OPEN (NO SHIELDS OR SEALS)
(CODE A)

BEARING CONFIGURATIONS

AMSC/NA

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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size code in table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields and Seals. All angular contact bearings with counterbored outer rings shall be open (no shields or seals, see figure 1) and designated by the code "A" in the part number.

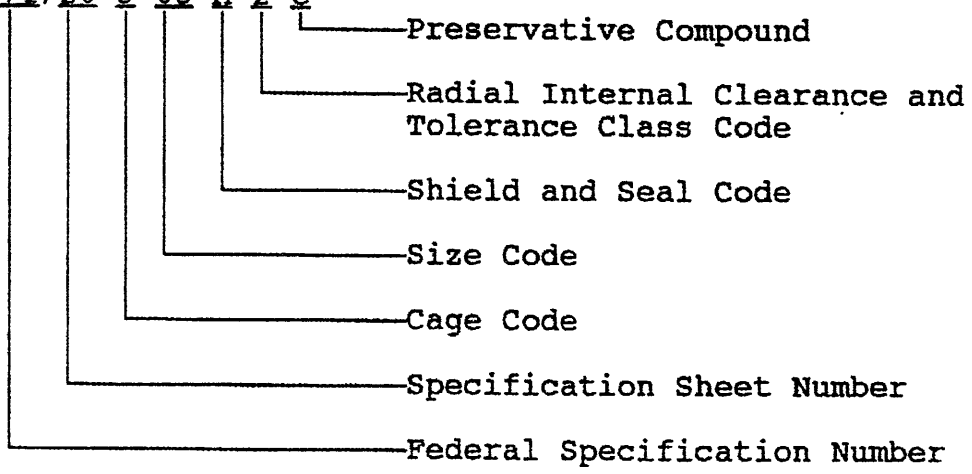
4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

5. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

6. Preservative Compound. All bearings shall be supplied with preservative compound in accordance with MIL-P-197 and designated by the code "C" in the part number.

7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields and seals, radial internal clearance and tolerance class, and preservative compound.

EXAMPLE: FFB171/20 J 03 A 2 C



FFB171/20J03A2C indicates - Bearing, cage - pressed steel or molded plastic: bore 17mm; outside diameter 35mm; width 10mm; open (no shields or seals); radial internal clearance symbol 0 (normal); tolerance class ABEC-1; preservative compound in accordance with MIL-P-197.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic Load Rating (Pounds)
00	10	26	8	0.3	12.5	795
01	12	28	8	0.3	14.5	870
02	15	32	9	0.3	17.5	940
03	17	35	10	0.3	19.5	1320
04	20	42	12	0.6	25	1680
05	25	47	12	0.6	30	1865
06	30	55	13	1.0	36	2410
07	35	62	14	1.0	41	2910
08	40	68	15	1.0	46	3000
09	45	75	16	1.0	51	3715
10	50	80	16	1.0	56	3820
11	55	90	18	1.1	62	5205
12	60	95	18	1.1	67	5350
13	65	100	18	1.1	72	5480
14	70	110	20	1.1	77	6470
15	75	115	20	1.1	82	6610
16	80	125	22	1.1	87	8500
17	85	130	22	1.1	92	8690
18	90	140	24	1.5	98.5	10515
19	95	145	24	1.5	103.5	12450
20	100	150	24	1.5	108.5	10990
21	105	160	26	2.0	115	13020
22	110	170	28	2.0	120	15100
24	120	180	28	2.0	130	15370
26	130	200	33	2.0	140	21055
28	140	210	33	2.0	150	20815
30	150	225	35	2.1	162	23810
32	160	240	38	2.1	172	26980
34	170	360	42	2.1	182	32310
36	180	280	46	2.1	192	36035

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamfer $r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met Provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)
Y	Pressed brass

TABLE III. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (Less than normal)	ABEC-1
2	Symbol 0 (Normal)	
3	Symbol 3 (Greater than normal)	
4	Symbol 4 (Greater than symbol 3)	
5	Symbol 2 (Less than normal)	ABEC-5
6	Symbol 0 (Normal)	
7	Symbol 3 (Greater than normal)	
8	Symbol 4 (Greater than symbol 3)	

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

(Project 3110-0848-20)

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